

Study Report: Agile Testing - QA & Test Engineer

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Family: QA & Test Engineer
Level: Intermediate
Chapters: 7
Source: Reference: Agile Testing - Lisa Crispin & Janet Gregory

Official sources and free libraries

- <https://agiletester.ca/agile-testing-book/>

Summary

Study report for **Agile Testing** by Lisa Crispin & Janet Gregory (Intermediate level) mapped to the QA & Test Engineer role. Testing in Scrum: whole-team quality and automation strategy.

Chapters

Track: Book-Intro

Chapter 1: About Agile Testing [Intermediate]

Chapter focus: About Agile Testing** **(Level: Intermediate)**

This study report summarizes **Agile Testing** by Lisa Crispin & Janet Gregory for the QA & Test Engineer role. The resource is rated Intermediate level. Testing in Scrum: whole-team quality and automation strategy. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Agile Testing
# Author: Lisa Crispin & Janet Gregory
# Role: QA & Test Engineer
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Agile Testing.

When to use it

When learning QA & Test Engineer skills at Intermediate level.

Where to use it

QA & Test Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role *(Level: Intermediate)**

QA & Test Engineer professionals use ideas from Agile Testing to solve real workplace problems. Testing in Scrum: whole-team quality and automation strategy. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: QA & Test Engineer

Book focus: Testing in Scrum: whole-team quality and automation strategy.

Recommended level: Intermediate

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

QA & Test Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered *(Level: Intermediate)**

The main topics in Agile Testing include practical concepts described as: Testing in Scrum: whole-team quality and automation strategy. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in QA & Test Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide**What it actually is**

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: pytest Automation Framework Structure [Intermediate]**Chapter focus: pytest Automation Framework Structure** *(Level: Intermediate)**

conftest.py shares fixtures across test packages without import boilerplate. Arrange-Act-Assert keeps tests readable; one logical assertion focus per test when possible. Markers (@pytest.mark.smoke) slice suites for fast PR gates vs full nightly runs. pytest.param combines inputs with test ids for clear failure output.

Code Reference:

```
import pytest

@pytest.mark.parametrize('email,expected', [
    ('user@example.com', 201),
    ('not-an-email', 422),
    ('', 422),
], ids=['valid', 'invalid-format', 'empty'])
def test_register_status(api_client, email, expected):
```

```
assert api_client.post('/register', json={'email': email}).status_code == expected
```

What it shows: parametrize runs three cases; ids label each in failure reports.

Topic guide

What it actually is

pytest is the standard Python test runner for API and unit automation.

When to use it

Adopt for all Python-based test automation in backend-heavy teams.

Where to use it

CI smoke suites, microservice contract tests, and data pipeline validation.

Real use example

Smoke marker runs in 3 minutes on PR; full regression runs nightly in 45 minutes.

Chapter 5: Applied: Selenium WebDriver and Page Object Model [Intermediate]

Chapter focus: Selenium WebDriver and Page Object Model** *(Level: Intermediate)

Selenium drives browsers programmatically; WebDriver W3C standardizes API across languages. Page Object Model encapsulates locators and actions in page classes reducing duplication. Explicit waits (WebDriverWait) handle async UI better than sleep(5) flakes. Run headless in CI; headed locally for debugging selector issues.

Code Reference:

```
class LoginPage:
    def __init__(self, driver):
        self.driver = driver
    def login(self, email, password):
        self.driver.find_element(By.ID, 'email').send_keys(email)
        self.driver.find_element(By.ID, 'password').send_keys(password)
        self.driver.find_element(By.CSS_SELECTOR, 'button[type=submit]').click()
```

What it shows: LoginPage hides locator details; tests call login() with credentials only.

Topic guide

What it actually is

Selenium automates browser UI regression at scale with mature ecosystem.

When to use it

Use when Playwright not available or legacy Java/C# grid infrastructure exists.

Where to use it

Enterprise regression grids, cross-browser corporate apps, and SAP web UIs.

Real use example

Locator change updates LoginPage once; fifty tests keep passing unchanged.

Chapter 6: Applied: Playwright Modern E2E Testing [Intermediate]

Chapter focus: Playwright Modern E2E Testing** *(Level: Intermediate)

Playwright auto-waits for element actionability reducing flaky explicit wait code. Trace viewer records DOM, network, and screenshots step-by-step on failure. Built-in test generator records actions producing starter locators-refine to getByRole. Python sync API suits pytest teams; async available for advanced flows.

Code Reference:

```
def test_search_product(page):
    page.goto('https://demo.shop')
    page.get_by_label('Search').fill('keyboard')
    page.get_by_role('button', name='Search').click()
    expect(page.get_by_role('heading', name='Results')).to_be_visible()
```

What it shows: get_by_label and get_by_role prefer accessible selectors over brittle CSS.

Topic guide

What it actually is

Playwright is the preferred 2024-2026 browser automation for new E2E suites.

When to use it

Start new E2E projects with Playwright unless constrained by existing Selenium grid.

Where to use it

Cross-browser CI, visual debug, and mobile viewport emulation.

Real use example

Failed CI uploads trace.zip; QA replays exact click sequence seeing network 500.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice** *(Level: Intermediate)

Finish Agile Testing with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to QA & Test Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
```

Week 4: Review + quiz

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a intermediate project screenshot.